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Efficient, portabLe And Secure orchesTration for reliable servICes

TrustMee: Self-Verifying Remote Attestation Evidence

2026-04-07

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The problem

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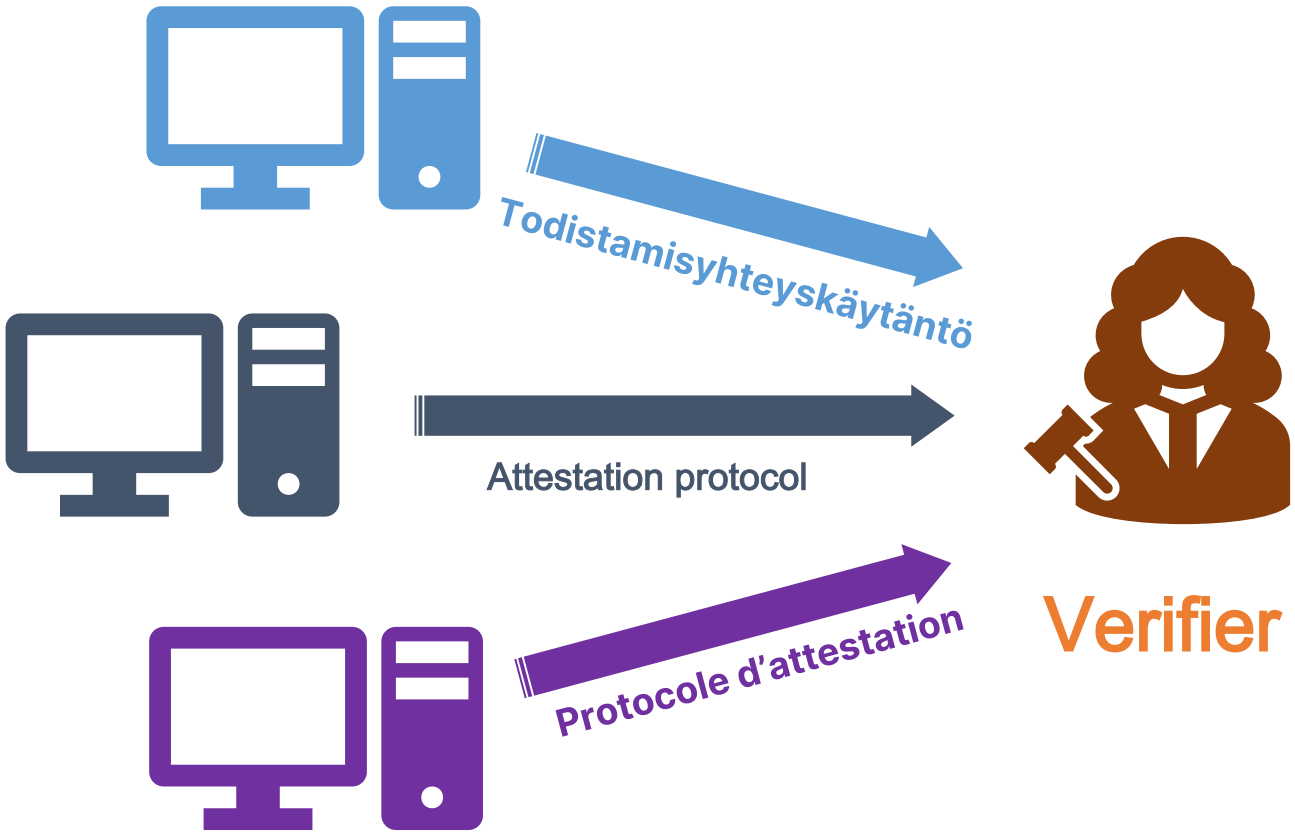

Attester

HW = Samsung A52
OS = Android 11
App = Bank ID

I'm talking to…
HW = Samsung A52
OS = Android 11
App = Bank ID

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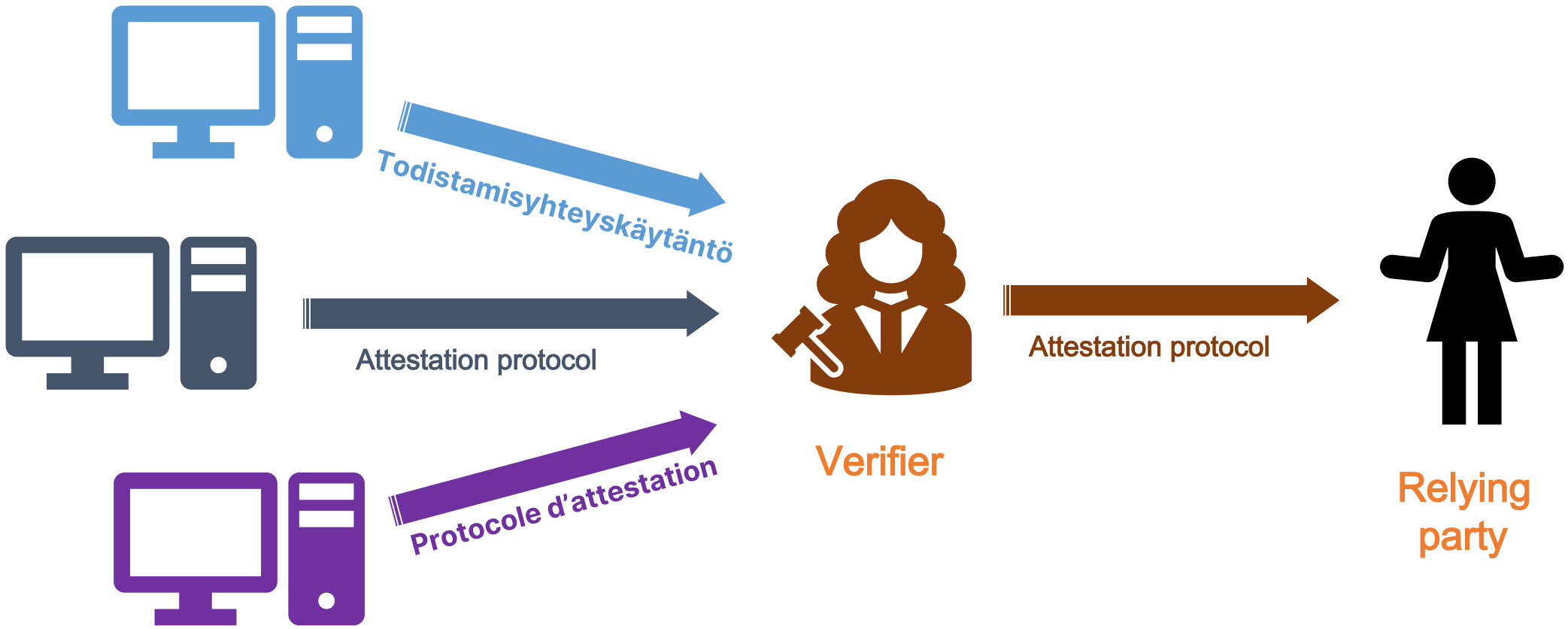
Verification frameworks



Solution: TrustMee

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Idea: Attester provides verification code to verifier

- Eliminates all platform-specific code from the verifier

Problem: Malicious verification code can compromise the verifier

Solution: **Protect the verifier using sandboxing techniques from the web**

Problem: How can we trust the verification logic

Solution: **Measure the verification code, now it's just another claim to check**

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WebAssembly: An application virtual machine

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Applications compiled into WebAssembly modules

Modules have (non-exhaustive)

- An isolated memory space
- Exports: functionality exposed to the VMM (e.g. program entry point)
- Imports: functionality requested from the VMM (e.g. storage operations)

Wasm VMs are cheap!

- We can put different parts of an application each in their own sandbox

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WebAssembly component model

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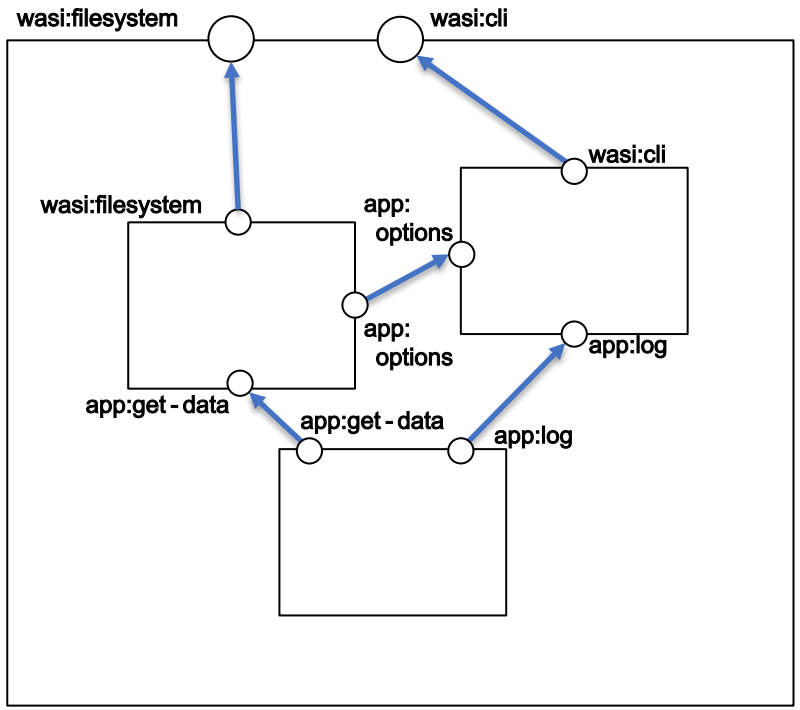
Wasm modules can be wrapped into Components with a well-defined ABI

- Wasm components can be composed into larger components




```
package trustee:verifier;

interface verifier-interface {
  variant optional-data {
    value(list<u8>),
    not-provided,
  }

  type report-data = optional-data;
  type init-data-hash = optional-data;

  resource verifier {
    constructor();
    evaluate: func(
      evidence: list<u8>,
      expected-report-data: report-data,
      expected-init-data-hash: init-data-hash,
    ) -> string;
  }
}

world verifier {
  export verifier-interface;
}
```

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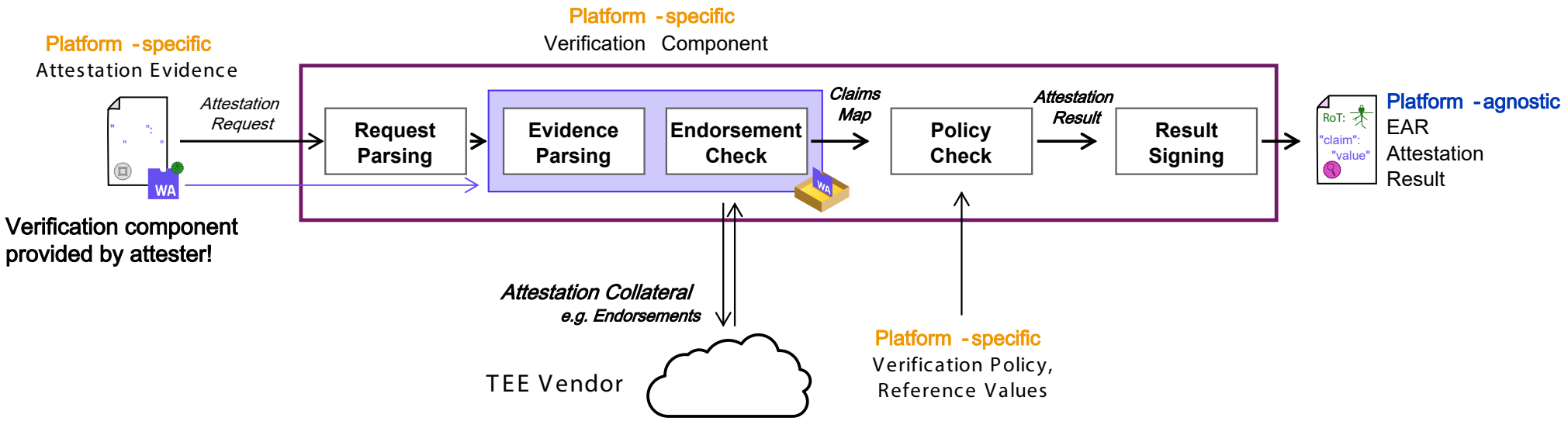
TrustMee



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TrustMee implementation

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Verification component in a Wasm sandbox

- Isolated filesystem region for caching, etc.

Optional network access for OCSP, fetching collateral, etc.

- Only for signed components (for now) to prevent network access from being abused

Implemented as platform driver in Trustee

- EAT+CMW wrapper for platform-specific evidence


```
{
    "tee": "tdx",
    "verifier": "wasm-verification-component",
    "evidence_bundle": {
        "eat_profile": "https://ssg.aalto.fi/trustmee/eat/component-evidence",
        "component_id": "component-<sha256 of-verifier-wasm-bytes>",
        "evidence_type": "application/octet-stream",
        "evidence": "<base64url(raw-evidence-bytes)>"
    }
}
```


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TrustMee implementation

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Two implementations, one public

Cross-compatibility tests

Verification components

- AMD SNP
- Intel TDX (Azure)
- AMD SNP again
- Intel TDX (Standalone)

Open-Source

Platform driver implementations

- Non-public prototype

github.com/elasticproject-eu/trustmee-lib

Open-Source

TrustMee implementation

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Average end-to-end attestation latency for SNP:

	Trustee native	TrustMee*	TrustMee with host crypto**
E2E attestation latency	5.3 ms	14.1 ms	6.5 ms
Platform-specific verification latency	1.1 ms	9.5 ms	2.7 ms

* Warm-start, with compiled verification component cached in memory

** Cryptography operations are performed on the host (in future wasi-crypto should replace it)

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The Future

WIP: Eliminating component certification

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Problem: Verification components isolated, but currently still require certification

- Rogue components could cause DoS (infinite loops, etc.)
- Components needing network access (e.g. for OCSP) potentially risky

Solution: Tighter restrictions still on the component

- Limits on computation time
- Limits on network access for uncertified components, examples:
  - Attester needs to staple collateral
  - Attester needs to proxy the request
  - Grant permission to access domains with special DNS entries
  - Limits on data transfer to prevent reflection attacks

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Layered & composite attestation

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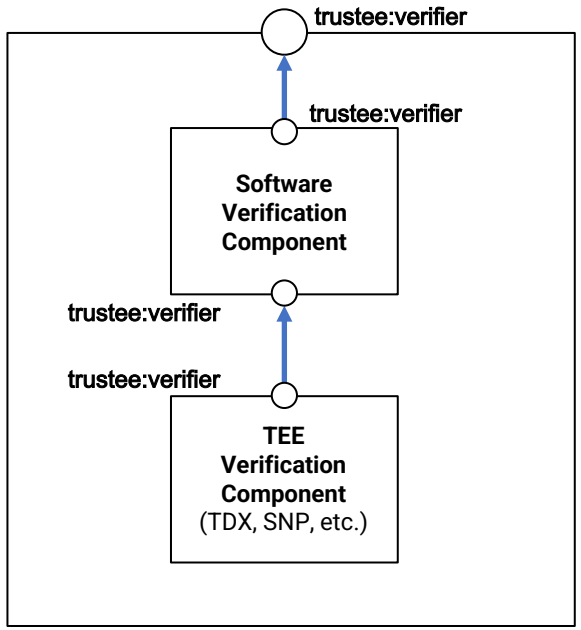
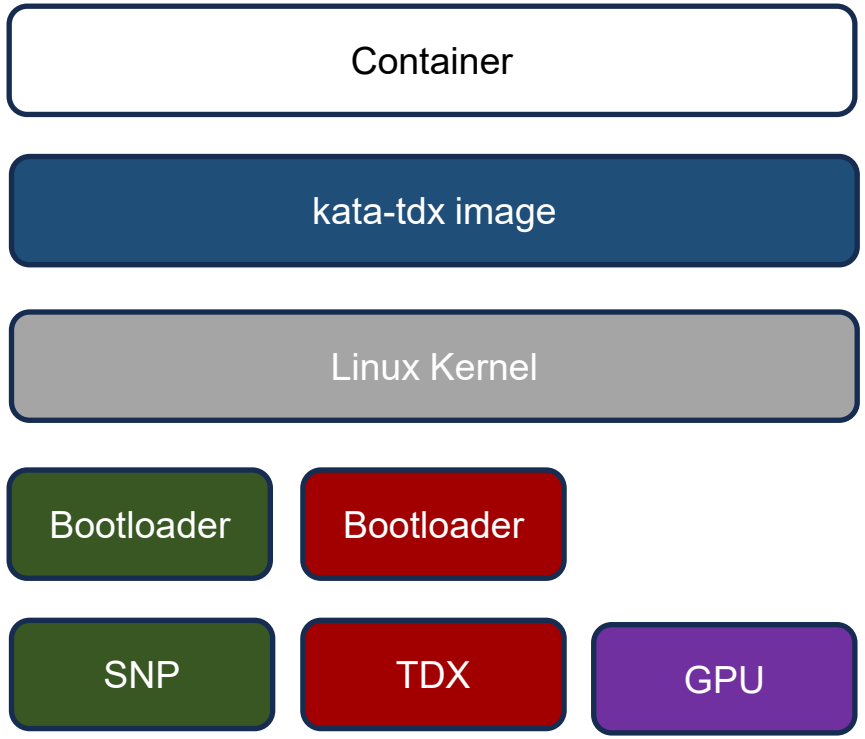
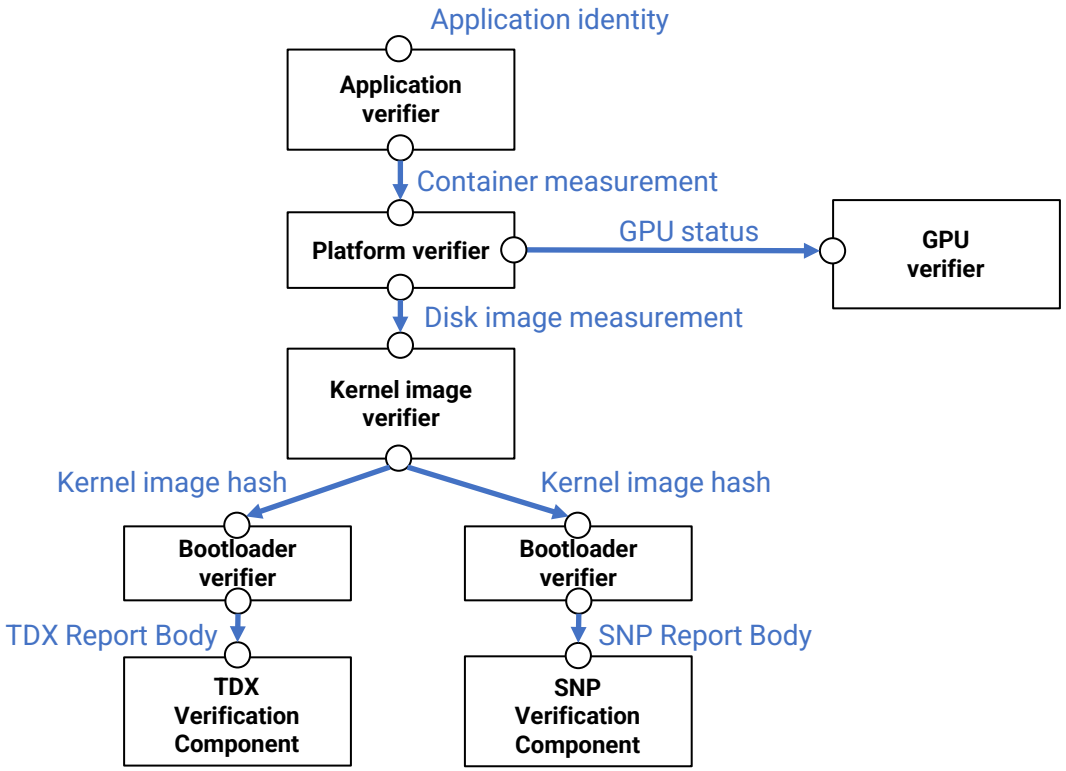
Problem: **Attestation doesn't come just from hardware, but a whole software stack too**

Solution: **Build TrustMee verification components for higher-level application platforms**



Example: container platform





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Example: distributed attestation

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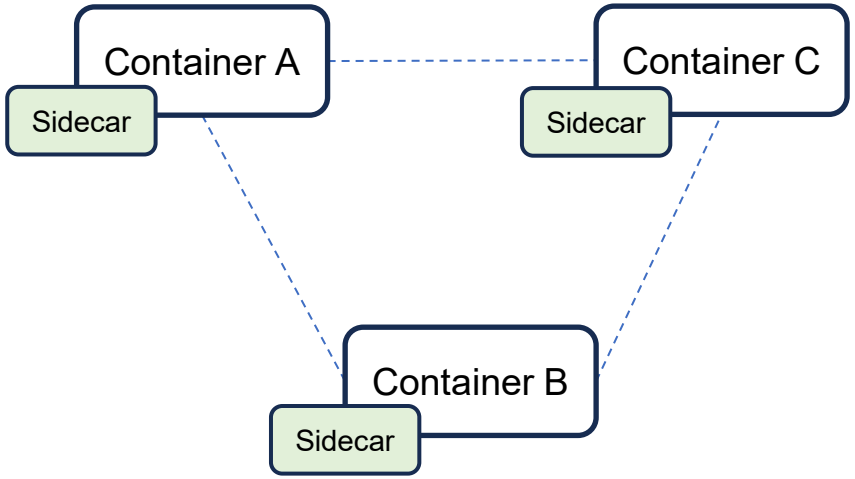
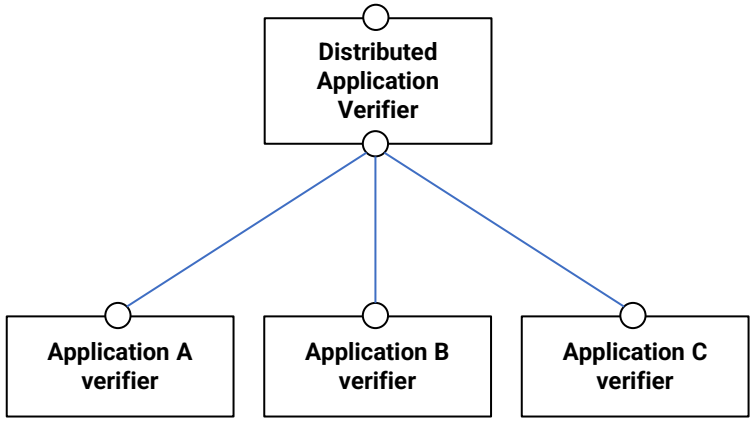
Consensus-like protocol combines component attestations into whole-app attestations





Immediate questions

Encapsulation

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How should we encapsulate the evidence?

- TrustMee evidence as an EAT token
  - TrustMee-specific EAT profile wrapping platform-specific evidence **Current implementation**
  - Platform-specific EAT profile, TrustMee metadata in JOSE wrapper
    - TrustMee = signature algorithm
    - evidence = signature
    - verification component = public key
    - Allows any EAT profile to self-authenticate without transformation by vendor's attestation service
- TrustMee evidence only as a CMW collection type, don't try to force non-EAT evidence into EAT
  - CMW needed anyway for stapled endorsements

How should we encapsulate the results (as a submod in the EAR document)?

- TrustMee as lead attester, with the verification component producing a nested token
  - Inserts extra layers of nested EAT tokens
  - No profile-compliance issues, since the nested token doesn't have extra claims not in its profile
- TrustMee claims inserted into existing output **Current implementation**
  - Resulting output doesn't comply with any EAT profile

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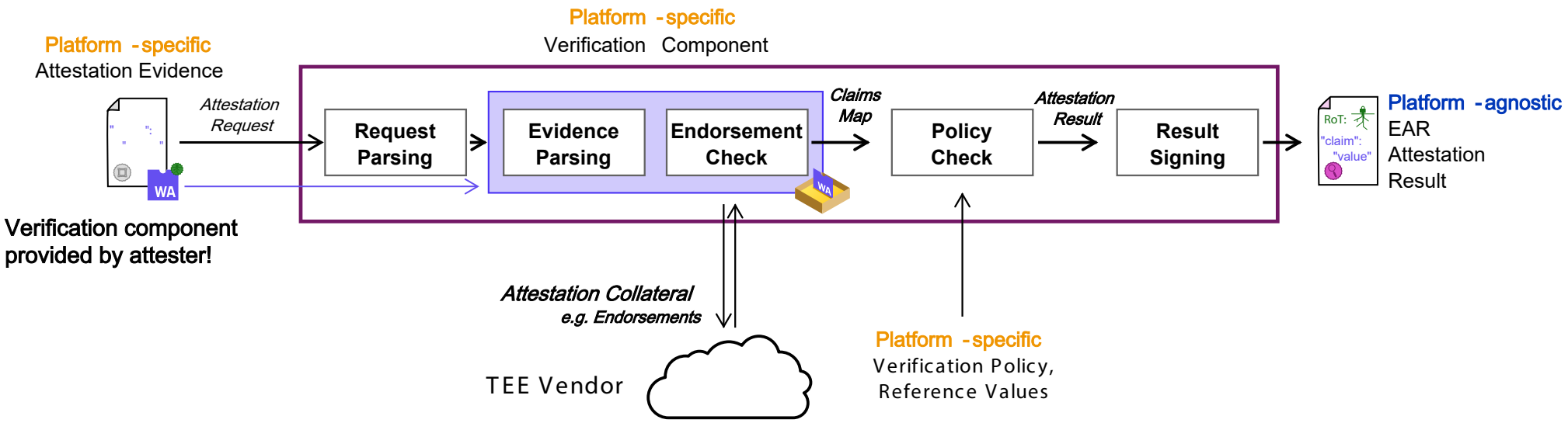
Reference Values

What's the best way to deal with reference values?

- Not currently addressed in TrustMee
- Should verifier components be able to provide or validate reference values themselves?



Summary

TrustMee allows attestation verifiers to eliminate platform-dependent logic

Platform-specific logic can be shared between different verifiers

Prototype available as open-source

- Standalone verification library
- Trustee platform driver

arxiv.org/abs/2602.13148

github .com/elasticproject - eu/trustmee - lib

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